

Real-Time Data Analyst Coding Challenge

Role: Junior Data Analyst

Healthcare Patient Analytics

Scenario

A hospital wants to reduce patient readmission rates.

Dataset Link: <https://www.kaggle.com/c/1056lab-diabetes-readmission-prediction>

Dataset Contains

- Patient ID
- Age
- Disease
- Admission Date
- Readmission
- Treatment Cost
- Doctor Assigned
- Recovery Days

Problem Statement

Identify factors causing frequent patient readmissions.

Tasks

- Handle missing medical records
- Standardize disease names
- Analyze age vs readmission
- Find costly treatments

Expected Visualizations

- Readmission by disease
- Patient age distribution

- Recovery trend analysis
- Cost analysis dashboard

Challenge

Find:

- Which diseases have highest readmission?
- Which doctors achieve faster recovery?
- Is treatment cost linked to recovery?

Storytelling Requirement

Prepare a healthcare insights report for hospital management.